

Computer Science Internal Assessment Criterion A - Planning

Total Word Count : 476

The scenario :

Mrs. [REDACTED] is the Vice President of Human Resources at Microland Limited, a global IT and infrastructure services company. She oversees HR operations for the company, managing a team of 35 people across different regions.

In the last quarter of every year, departments submit budget requests for the next year and during our initial discussion, Mrs. Usha explained that the HR admin team, led by her, ensures that these budgets align with the company's financial growth strategy and operational requirements.

Currently, the budget-planning system relies on manual tracking via Excel sheets. Mrs. Usha stated that each department's budget is reviewed individually on a separate file, but the process has proven to be inefficient and time-consuming due to errors in tracking, data discrepancies, and a lack of data security.

The client's primary concern with manual tracking is the low security and data disorganisation, as she highlighted that the current system is prone to errors like incorrect tagging of expenses or missed reviews due to lack of ability to monitor real-time spending. Additionally, it's difficult to control access to sensitive budget data as multiple entities are involved in the budget approval-and-tracking process. Hence, there's no mechanism to ensure departments are making appropriate purchasing and budget decisions.

Note: this section was written based on consultation with the client, refer to the appendix A for the transcript of the first interview

Scenario word count: 205

Rationale for the proposed solution:

To address these issues, we discussed and agreed that a centralised web application would ensure a secure, organised, and transparent budget-management process. This allows departments and leadership to submit budget requests, track planned-versus-actual expenditure in real time, and access approved budget plans. The system would provide role-based-access control, ensuring only authorised users can view and edit sensitive data.¹

¹ ORACLE. "What Is MySQL?" *Oracle.com*, 2023, www.oracle.com/in/mysql/what-is-mysql/.

A web application instead of a mobile app is ideal as Mrs. Usha handles multiple regional teams, and the platform can be conveniently accessed across devices. The interface will employ HTML and CSS for a clean layout, with JavaScript for interactivity.

I chose Node.js for the backend because its asynchronous, event-driven programming model can handle multiple user requests concurrently.² The client agreed that this allows the HR team to manage submissions from multiple employees without lags. Additionally, Node.js is cost-effective, scalable and has built-in error handling to prevent approval delays. It also has strict validation processes for budget inputs, reducing data discrepancies - one of the client's key concerns.^{3 4}

The MySQL database securely stores department-wise budget data and tracks planned versus actual spending in real time. Its ACID compliance ensures reliable access and consistency, while structured queries allow accurate comparisons between requested budgets and expenditures.

Using a MySQL database would securely store department-wise budget data and enable organised expenditure-tracking with time-stamps. The software's ACID compliance ensures reliable and consistent access, and structured queries allow the HR team to generate precise comparisons between requested and actual budgets, while reducing data redundancy and increasing security.

Overall, the application streamlines budget tracking, enhances transparency, and minimizes errors through real-time data consistency and automated approval workflows.

Note: this section was written based on consultation with the client, refer to the appendix A for the transcript of the second interview

Rationale word count : 271

² GeeksforGeeks. "Asynchronous Programming in NodeJS." *GeeksforGeeks*, 17 July 2020, www.geeksforgeeks.org/node-js/asynchronous-patterns-in-node-js/. Accessed 10 Sept. 2025.

³ Ciszewski, Bartek. "Maximizing Node.js Scalability: Best Practices, Tools, and Patterns." *Netguru.com*, Netguru, 2023, www.netguru.com/blog/node-js-scalability.

⁴ "The Pros and Cons of Node.JS in Web Development." *GeeksforGeeks*, 5 Dec. 2022, www.geeksforgeeks.org/node-js/the-pros-and-cons-of-node-js-in-web-development/.

Success Criteria

- A. Authentication features that provide role-based access control (Admin, Department Head, General Employee).
- B. Implement session management to maintain active user sessions securely, ensuring users remain authenticated while performing actions within the system.
- C. All passwords should be hashed.
- D. Users (all users) should be able to log in and reset their passwords.
- E. Admin should be able to view and compare budget plans with actual budget implementations for each department in graphical form.
- F. Admin should be able to search for and filter employees based on department name and role.
- G. Admin should have a clear overview of budget data and status for each department.
- H. Admin should be able to create, edit, and delete user accounts.
- I. Admin should be able to edit departmental budget requests and budget status.
- J. Department Heads should be able to edit budget plans for their department on their portal.
- K. Department Heads should be able to submit expenditure requests to admins for approval.
- L. Department Heads should be able to track their submitted requests and view approval status.
- M. Admin should be able to review and comment on budget adjustments.
- N. Admin should be able to track departmental expenditure patterns.
- O. Admin should be able to generate, export, and email comprehensive departmental budget reports.

- P. Users should be able to view all budget data relevant to their assigned role.
- Q. Department Heads should be able to add expenditures to a spending table whenever a purchase is made.
- R. Admin should have real-time budget tracking with email notifications of important departmental expenditures.
- S. New users should receive an onboarding guide explaining system functionalities.
- T. Each role should have a customized dashboard displaying relevant financial data and updates.
- U. General Employees and Department Heads should receive an email notification whenever a budget is approved or rejected by admin.
- V. Admin should receive an email notification whenever a budget request is submitted and pending approval.

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